nature medicine

Supplementary information https://doi.org/10.1038/s41591-025-03629-3

Changing climate and socioeconomic factors contribute to global antimicrobial resistance

In the format provided by the authors and unedited

Supplemental material

Table S1-1 Surveillance records of AMR profiles

Resistant profiles	Countries	Records	Records from high-income groups (%)	Time span	Isolates
3GCREC	96	945	637 (67.4%)	1999~2022	8527728
3GCRKP	96	805	535 (66.4%)	1999~2022	4973588
CREC	95	870	594 (68.3%)	1999~2022	8412847
CRKP	96	774	524 (67.7%)	1999~2022	4968993
CRAB	79	510	306 (60.0%)	1999~2022	2185560
CRPA	52	598	466 (77.9%)	1999~2022	2844380
Overall	101	4502	2425 (53.9%)	1999~2022	31913096

4 **Table S1-2** Surveillance records of AMR profiles in 101 countries

Countries (n = 101)	*3GCREC (n = 945)*	*3GCRKP (n = 805)*	*CRAB (n = 510)*	*CREC (n = 870)*	*CRKP (n = 774)*	*CRPA (n = 598)*	*Records (n = 4502)*	*Data sources*
Argentina	2014~2020	2014~2020	2014~2020	2014~2020	2014~2020	2014~2017	39	WHONET-Argentina Network and SIREVA II Argentina Network, Global Antimicrobial Resistance Surveillance System (GLASS)
Australia	2013~2017, 2019, 2020	2013~2017, 2019, 2020	2016, 2017, 2019, 2020	2013~2017, 2019, 2020	2013~2017, 2019, 2020	2015~2017	35	Australian Group on Antimicrobial Resistance (AGAR), Global Antimicrobial Resistance Surveillance System (GLASS)
Austria	2001~2022	2005~2022	2013~2022	2001~2022	2005~2022	2005~2022	108	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Bahrain	2016, 2018~2020	2017~2020	2018~2020	2016, 2018~2020	2017~2020	No records	19	Global Antimicrobial Resistance Surveillance System (GLASS)
Bangladesh	2019	No records	No records	2019	No records	No records	2	Global Antimicrobial Resistance Surveillance System (GLASS)
Belarus	2012, 2014~2019	2012, 2014~2018	2012, 2014~2019	2014~2019	2012, 2014~2019	2014~2019	39	Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR), Central Asian and European Surveillance of Antimicrobial Resistance (CAESAR)
Belgium	2001~2022	2009~2022	2016~2022	2010~2022	2009~2022	2009~2022	84	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Bhutan	2019, 2020	2019	No records	2019, 2020	No records	No records	5	Global Antimicrobial Resistance Surveillance System (GLASS)

Bosnia and Herzegovina	2015~2020	2015~2020	2015~2020	2015~2020	2015~2020	2015~2019	35	Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR), Central Asian and European Surveillance of Antimicrobial Resistance (CAESAR), Global Antimicrobial Resistance Surveillance System (GLASS)
Brazil	2018, 2019	2018, 2019	2019	2018, 2019	2018, 2019	No records	9	Global Antimicrobial Resistance Surveillance System (GLASS)
Brunei	2019, 2020	2019, 2020	2019, 2020	2019, 2020	2019, 2020	No records	10	Global Antimicrobial Resistance Surveillance System (GLASS)
Bulgaria	2001~2022	2005, 2006, 2008~2022	2012~2022	2001~2022	2005, 2006, 2008~2022	2005, 2009~2022	104	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Burkina Faso	No records	No records	No records	No records	2020	No records	1	Global Antimicrobial Resistance Surveillance System (GLASS)
Cambodia	2018~2020	2018, 2020	2018~2020	2018~2020	2018, 2020	No records	13	Global Antimicrobial Resistance Surveillance System (GLASS)
Canada	2012~2014	2012~2014	No records	2012~2014	2012~2014	2012~2014	15	Canadian Antimicrobial Resistance Alliance (CARA)
Chile	2014	No records	No records	2014	No records	No records	2	Chilean Society of infectious diseases
China	2014~2022	2014~2022	2014~2022	2014~2022	2014~2022	2014~2022	54	China Antimicrobial Resistance Surveillance System (CARSS)
Croatia	2001~2008, 2010~2022	2005~2008, 2010~2022	2013~2022	2001~2008, 2010~2022	2005~2008, 2010~2022	2005~2008, 2010~2022	103	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Cyprus	2004~2022	2007~2022	2013~2015, 2017~2022	2004~2022	2007~2022	2006~2022	96	European Antimicrobial Resistance Surveillance Network (EARS-Net)

Czech Republic	2001~2022	2005~2022	2013~2022	2001, 2005~2022	2005, 2007~2022	2005~2022	104	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Denmark	2005~2022	2006~2022	2012~2022	2005~2022	2006~2022	2007~2022	97	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Ecuador	2014	2014	2014	2014	2014	2014	6	Reference Laboratory for antimicrobial resistance at the National Institute of Public Health Research (INSPI), Ecuador
Egypt	2018~2020	2016~2020	2018~2020	2018~2020	2016~2020	No records	19	Global Antimicrobial Resistance Surveillance System (GLASS)
Estonia	2001~2022	2005~2022	No records	2001~2022	2007~2022	2005~2008, 2010, 2012, 2014, 2016~2022	92	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Ethiopia	No records	2018~2020	2020	2020	2018~2020	No records	8	Global Antimicrobial Resistance Surveillance System (GLASS)
Finland	2001~2022	2005~2022	2013~2015, 2017, 2019~2021	2001~2022	2005~2022	2005~2022	105	European Antimicrobial Resistance Surveillance Network (EARS-Net)
France	2002~2022	2005~2022	2012~2022	2002~2022	2005~2022	2005~2022	107	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Georgia	2018~2020	2016, 2018~2020	2017~2020	2018~2020	2016, 2018~2020	2019	19	Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR), Central Asian and European Surveillance of Antimicrobial Resistance (CAESAR), Global Antimicrobial Resistance Surveillance

								System (GLASS)
Germany	2000~2022	2005~2022	2012~2022	2001~2022	2005~2022	2005~2022	110	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Ghana	2014~2016	2014~2016	2014~2016	2014~2016	2014~2016	2014~2016	18	MDS-Lancet Laboratories Ghana Limited
Greece	2000~2022	2005~2022	2012~2022	2000~2022	2005~2022	2005~2022	111	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Hungary	2001~2022	2005~2022	2012~2022	2001~2022	2005~2022	2005~2022	109	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Iceland	2001~2022	2013, 2015, 2020, 2022	No records	2001~2010, 2020~2022	2020, 2022	2021, 2022	43	European Antimicrobial Resistance Surveillance Network (EARS-Net)
India	2008~2020	2008~2020	2008~2020	2008~2020	2008~2020	2009~2017	74	Private laboratory network, India, Private laboratory network/Indian Council of Medical Research/Fortis Healthcare Limited, Max Healthcare Institute Limited, Indian Council of Medical Research, Global Antimicrobial Resistance Surveillance System (GLASS)
Indonesia	2018~2020	2018~2020	2018~2020	2018~2020	2018~2020	No records	15	Global Antimicrobial Resistance Surveillance System (GLASS)
Iran	2018~2020	2018~2020	2018~2020	2018~2020	2018~2020	No records	15	Global Antimicrobial Resistance Surveillance System (GLASS)
Iraq	2020	2020	2020	2020	2020	No records	5	Global Antimicrobial Resistance Surveillance System (GLASS)
Ireland	2002~2022	2005~2022	2013~2022	2002~2022	2006~2022	2006~2022	104	European Antimicrobial Resistance

								Surveillance Network (EARS-Net)
Italy	2002~2022	2005~2022	2012~2022	2007~2022	2006~2022	2006~2022	100	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Japan	2017~2020	2017~2020	2017~2020	2017~2020	2017~2020	No records	20	Global Antimicrobial Resistance Surveillance System (GLASS)
Jordan	2018~2020	2018~2020	2018~2020	2018~2020	2018~2020	No records	15	Global Antimicrobial Resistance Surveillance System (GLASS)
Kenya	2012, 2015	2012, 2015	No records	2012, 2015	2012, 2015	No records	8	A Private tertiary-care hospital, Kenya
Korea, Rep.	2016~2020	2016~2020	2016~2020	2016~2020	2016~2020	No records	25	Global Antimicrobial Resistance Surveillance System (GLASS)
Kosovo	2014	2014, 2016~2020	2014~2020	2014	2014, 2016~2020	No records	21	National Institute for Public Health of Kosovo, Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR), Global Antimicrobial Resistance Surveillance System (GLASS)
Kuwait	2020	2020	2020	2020	2020	No records	5	Global Antimicrobial Resistance Surveillance System (GLASS)
Laos	2018~2020	2018~2020	No records	2018~2020	2020	No records	10	Global Antimicrobial Resistance Surveillance System (GLASS)
Latvia	2006~2022	2008~2022	2014~2022	2006~2022	2008~2022	2018~2022	78	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Lebanon	2016, 2018~2020	2018~2020	2018~2020	2016, 2018~2020	2018~2020	No records	17	Global Antimicrobial Resistance Surveillance System (GLASS)
Liberia	2020	2020	No records	No records	2020	No records	3	Global Antimicrobial Resistance Surveillance

								System (GLASS)
Libya	No records	2019, 2020	No records	No records	2019	No records	3	Global Antimicrobial Resistance Surveillance System (GLASS)
Lithuania	2006~2022	2006~2022	2014~2022	2006~2010, 2012~2022	2006~2010, 2012~2022	2010, 2011, 2013~2022	87	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Luxembourg	2001~2022	2007, 2008, 2010~2022	No records	2001~2005, 2007, 2009~2022	2010~2022	2009~2014, 2016~2022	83	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Macedonia, FYR	2013~2016, 2018~2020	2013, 2014, 2018~2020	2015, 2016, 2019, 2020	2013~2016, 2018~2020	2013, 2014, 2018~2020	No records	28	Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR), Global Antimicrobial Resistance Surveillance System (GLASS)
Madagascar	2019	2018, 2019	No records	2019	2018, 2019	No records	6	Global Antimicrobial Resistance Surveillance System (GLASS)
Malawi	2003~2014, 2016, 2017, 2020	2004~2015, 2017, 2020	2016, 2017, 2020	2016, 2017, 2020	2016, 2017, 2020	No records	38	Malawi Liverpool Wellcome Trust Clinical Research Programme, University of Malawi the College of Medicine, Global Antimicrobial Resistance Surveillance System (GLASS)
Malaysia	2013~2015, 2017~2020	2013~2015, 2017~2020	2013~2015, 2017~2020	2013~2015, 2017~2020	2013~2015, 2017~2020	No records	35	Ministry of Health, Malaysia, Global Antimicrobial Resistance Surveillance System (GLASS)
Mali	2018~2020	2018, 2020	No records	No records	No records	No records	5	Global Antimicrobial Resistance Surveillance System (GLASS)
Malta	2001~2022	2006,	No records	2001~2022	2006,	2005~2012,	91	European Antimicrobial Resistance

		2008~2022			2008~2022	2014, 2016, 2017, 2019~2022		Surveillance Network (EARS-Net)
Mauritius	2020	2020	2020	2020	2020	No records	5	Global Antimicrobial Resistance Surveillance System (GLASS)
Mexico	2005~2012, 2014, 2015	2005~2008, 2010, 2011, 2014, 2015	2007~2014	2006~2015	2005~2008, 2010~2015	2006~2015	56	Hospital Civil de Guadalajara Fray Antonio Alcalde
Moldova	No records	2019, 2020	2020	No records	2019, 2020	No records	5	Global Antimicrobial Resistance Surveillance System (GLASS)
Montenegro	No records	No records	2019	No records	No records	No records	1	Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR)
Myanmar	2018~2020	2019, 2020	2019, 2020	2018~2020	2019, 2020	No records	12	Global Antimicrobial Resistance Surveillance System (GLASS)
Nepal	2018~2020	2018~2020	2018~2020	2018~2020	2018~2020	No records	15	Global Antimicrobial Resistance Surveillance System (GLASS)
Netherlands	2000~2022	2005~2022	2012~2022	2000~2022	2005~2022	2005~2022	111	European Antimicrobial Resistance Surveillance Network (EARS-Net)
New Zealand	2002~2015	2004~2015	No records	2002~2015	2004~2015	No records	52	Public Health Surveillance, Institute of Environmental Science and Research Ltd (ESR), New Zealand
Nigeria	2017, 2018, 2020	2017, 2018, 2020	No records	2017	2018, 2020	No records	9	Global Antimicrobial Resistance Surveillance System (GLASS)
Norway	2000~2022	2005~2022	2013~2018, 2020~2022	2000~2022	2005~2022	2005~2022	109	European Antimicrobial Resistance Surveillance Network (EARS-Net)

Oman	2017~2020	2017~2020	2017~2020	2017~2020	2017~2020	No records	20	Global Antimicrobial Resistance Surveillance System (GLASS)
Pakistan	2011~2015, 2017~2020	2014, 2015, 2017~2020	2017~2020	2011~2015, 2017~2020	2014, 2015, 2017~2020	No records	34	Chughtai's laboratory, Lahore (CLL), Global Antimicrobial Resistance Surveillance System (GLASS)
Palestinian Territory	2020	2020	2020	2020	2020	No records	5	Global Antimicrobial Resistance Surveillance System (GLASS)
Peru	2020	2020	2020	2020	2020	No records	5	Global Antimicrobial Resistance Surveillance System (GLASS)
Philippines	2016~2020	2016~2020	2016~2020	2016~2020	2016~2020	2016	26	Global Antimicrobial Resistance Surveillance System (GLASS)
Poland	2001~2022	2005~2007, 2009~2022	2012~2022	2009~2022	2005, 2009~2022	2006, 2007, 2009~2022	95	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Portugal	2001~2022	2006~2022	2012~2022	2007~2022	2007~2022	2006~2022	99	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Qatar	2019, 2020	2019, 2020	No records	2019, 2020	2019, 2020	No records	8	Global Antimicrobial Resistance Surveillance System (GLASS)
Romania	2003~2022	2006, 2007, 2012~2022	2012~2022	2003~2008, 2010~2022	2006, 2012~2022	2012~2022	86	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Russia	2015, 2016, 2018~2020	2014~2016, 2018~2020	2014~2016, 2018~2020	2015, 2016, 2018~2020	2014~2016, 2018~2020	2016~2019	32	Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR), Global Antimicrobial Resistance Surveillance System (GLASS)
Saudi Arabia	2017~2020	2017~2020	2017~2020	2017~2020	2017~2020	No records	20	Global Antimicrobial Resistance Surveillance System (GLASS)

Serbia	2013~2019	2013~2019	2013~2019	2013~2019	2013~2019	2013~2019	42	Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR), Central Asian and European Surveillance of Antimicrobial Resistance (CAESAR)
Singapore	2019, 2020	2019, 2020	No records	2019, 2020	2019, 2020	No records	8	Global Antimicrobial Resistance Surveillance System (GLASS)
Slovakia	2001~2005, 2011~2022	2011~2022	2013~2022	2002~2005, 2011~2022	2011~2022	2011~2022	79	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Slovenia	2001~2022	2005~2022	2014~2022	2002~2022	2005~2022	2005~2022	106	European Antimicrobial Resistance Surveillance Network (EARS-Net)
South Africa	2011~2016, 2018~2020	2011~2016, 2018~2020	2011~2016, 2018~2020	2011~2016, 2018~2020	2011~2016, 2018~2020	2011~2016	51	SA National Health Laboratory Services and SASCM Private Data, South Africa, Global Antimicrobial Resistance Surveillance System (GLASS)
Spain	2001~2022	2005~2022	2013~2022	2001~2022	2005~2022	2005~2022	108	European Antimicrobial Resistance Surveillance Network (EARS-Net)
Sri Lanka	2009, 2019, 2020	2009, 2019, 2020	2009, 2019, 2020	2009, 2019, 2020	2009, 2019, 2020	No records	15	The Sri Lanka College of Microbiologists, Global Antimicrobial Resistance Surveillance System (GLASS)
Sudan	2018	2018	No records	2018	2018	No records	4	Global Antimicrobial Resistance Surveillance System (GLASS)
Sweden	2001~2008, 2010~2022	2005~2008, 2010~2022	2013~2022	2005, 2007, 2008, 2010~2022	2007, 2008, 2010~2022	2005~2008, 2010~2022	96	European Antimicrobial Resistance Surveillance Network (EARS-Net)

Switzerland	2013~2016, 2018~2020	2013~2016, 2018~2020	2013~2016, 2018~2020	2013~2016, 2018~2020	2013~2016, 2018~2020	2013~2019	42	Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR), Global Antimicrobial Resistance Surveillance System (GLASS)
Taiwan	2007~2016	2007~2016	2007~2016	2007~2016	2007~2016	2007~2016	60	Taiwan National Institutes of Infectious Diseases and Vaccinology
Thailand	2008~2020	2008~2020	2008~2015, 2017~2020	2008~2020	2008~2015, 2017~2020	2008~2015	71	National Antimicrobial Resistance Surveillance Center, Thailand (NARST), Global Antimicrobial Resistance Surveillance System (GLASS)
Tunisia	2016~2020	2016~2020	2016~2020	2016~2020	2016~2020	No records	25	Global Antimicrobial Resistance Surveillance System (GLASS)
Turkey	2013~2019	2013~2019	2014~2019	2013~2019	2013~2019	2013~2019	41	Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR), Central Asian and European Surveillance of Antimicrobial Resistance (CAESAR)
Ukraine	2019, 2020	2018~2020	2017, 2019, 2020	2019, 2020	2018~2020	No records	13	Central Asian and European Surveillance of Antimicrobial Resistance (CAESAR), Central Asian and Eastern European Surveillance of Antimicrobial Resistance (CAESAR), Global Antimicrobial Resistance Surveillance System (GLASS)
United Arab	2017~2020	2017~2020	2017~2020	2017~2020	2017~2020	No records	20	Global Antimicrobial Resistance Surveillance

Emirates								System (GLASS)
United Kingdom	2001~2020	2005~2020	2012~2020	2006~2020	2005~2020	2005~2019	91	European Antimicrobial Resistance Surveillance Network (EARS-Net), Global Antimicrobial Resistance Surveillance System (GLASS)
United States	1999~2016	1999~2016	1999~2016	1999~2016	1999~2016	1999~2016	108	The Surveillance Network, USA, The Surveillance Network, USA/ResistanceMap Surveillance Network, National Healthcare Safety Network, USA/ResistanceMap Surveillance Network, Becton Dickinson
Venezuela	2000~2013	2000~2013	2000~2012	2000~2013	2000~2013	2000~2013	83	Programa Venezolano de Vigilancia de la Resistencia a los Antimicrobianos (PROVENRA), Venezuela
Vietnam	2013, 2016	2013, 2016	2013, 2016	2013, 2016	2013, 2016	2013, 2016	12	Vietnam Resistance Project (VINARES)
Zambia	2019	2016, 2017, 2019	No records	2019	2016, 2019	No records	7	Global Antimicrobial Resistance Surveillance System (GLASS)
Zimbabwe	2015	No records	No records	2015	2015	No records	3	Lancet Laboratories

6 **Table S2** Description of independent variables

Variables	Description	Measurement unit	Observational source	Predicational source
$PM_{2.5}$	Fine particulate matter smaller than 2.5 micrometres, including dust, dirt, soot, and liquid droplets. Particles of this size is capable of penetrating deep into the lung and enter the bloodstream causing cardiovascular (ischaemic heart disease), cerebrovascular (stroke) and respiratory impacts.	$\mu g/m^3$	ACAG	CMIP6
Surface runoff	The flow of water over the ground surface, often resulting from precipitation or melting snow.	100mm	ERA5	CMIP6
Sub-surface runoff	The flow of water beneath the ground surface, often resulting from precipitation or melting snow.	100mm	ERA5	CMIP6
Temperature change on land	The change in near-ground ambient temperature relative to a fixed baseline: the average near-ground ambient temperature from 1951 to 1980.	℃	FAOSTAT	CMIP6
Urban population	People living in urban areas as defined by national statistical offices.	% of total population	World Bank	SSP Public Database
Corruption Perceptions Index	A composite index ranking countries and territories based on the perceived levels of public sector corruption according to experts and businesspeople.	Index (1-100)	Transparency International	Linear prediction
Health expenditure	Expenditure on health per capita expressed in international dollars at purchasing power parity.	% of GDP	World Bank	Linear prediction
Out-of-pocket expenditure	Health expenditure through out-of-pocket payments per capita in international dollars at purchasing power parity.	% of total health expenditure	World Bank	Linear prediction
Population density	Residents per square kilometre, regardless of legal status or citizenship.	log-scale transformed	World Bank	SSP Public Database

International tourists	The number of tourists who travel to a country other than that in which they usually reside.	log-scale transformed	World Bank	Linear prediction
Water, Sanitation, and Hygiene (WASH) coverage	The percentage of people using safe drinking-water, sanitation and hygiene.	% total population	World Bank	Linear prediction
Immunisation coverage	Complete child immunisation rate of polio, hepatitis B, BCG, DPT, Hib3, measles, and second dose of measles.	% total population	World Bank	Linear prediction
Antimicrobial consumption	Total consumption of antimicrobial products in a country, measured by defined daily dose (DDD).	DDD/1,000/day	GLASS, ESAC-Net, GRAM	Linear prediction
Income groups	Economies classified by the World Bank Atlas using gross national income (GNI) per capita (Figure S3).	/	World Bank	Consistent with 2022
J01 class consumption	Antimicrobial products consumption of different J01 class in a country, measured by defined daily dose (DDD).	DDD/1,000/day	GRAM	/
School enrollment	The ratio of total enrollment, regardless of age, to the population of the age group that officially corresponds to the level of education.	% school-age population	World Bank	/
Antimicrobials in animals	Antimicrobial agents intended for use in animals.	mg/kg	ANIMUSE	/

Note: BCG = *Bacillus Calmette-Guérin*. DPT = *Diphtheria, Pertussis, and Tetanus* Vaccine. Hib = *Haemophilus influenzae* type b.

8 **Table S3** Variables and data source

Variables	*Data source*	*URL*
Antimicrobial resistance	Antibiotic resistance from ResistanceMap	https://resistancemap.onehealthtrust.org/AntibioticResistance.php
	*Central Asian and European Surveillance of Antimicrobial Resistance (CAESAR) network	https://www.who.int/europe/publications/i/item/WHO-EURO-2020-3469-43228-60585
	European Antimicrobial Resistance Surveillance Network (EARS-Net)	https://www.ecdc.europa.eu/en/antimicrobial-resistance/surveillance-and-disease-data
	China Antimicrobial Resistance Surveillance System (CRASS)	https://www.carss.cn/
Temperature change on land	Food and Agriculture Organization Statistics (FAOSTAT)	https://www.fao.org/faostat/en/#data/ET
	Coupled Model Intercomparison Project Phase 6 (CMIP6)	https://aims2.llnl.gov/search/cmip6/
Geographic-mean $PM_{2.5}$	Atmospheric Composition Analysis Group (ACAG)	https://sites.wustl.edu/acag/datasets/surface-pm2-5/
	Coupled Model Intercomparison Project Phase 6 (CMIP6)	https://aims2.llnl.gov/search/cmip6/
Surface runoff, Subsurface runoff	European Centre for Medium-Range Weather Forecasts Reanalysis v5 (ERA5)	https://cds.climate.copernicus.eu/datasets
	Coupled Model Intercomparison Project Phase 6 (CMIP6)	https://aims2.llnl.gov/search/cmip6/
Urban population, Population per square kilometre	World Development Indicators	https://databank.worldbank.org/
	Shared Socioeconomic Pathways (SSP) Public Database	https://tntcat.iiasa.ac.at/SspDb/
Corruption Perceptions Index	Transparency International	https://us.transparency.org/
Health expenditures, out-of-pocket expenditures, international tourism, Water, Sanitation, and Hygiene (WASH) coverage, immunisation coverage, primary school enrolment, secondary school enrolment	World Development Indicators	https://databank.worldbank.org/

Antimicrobial consumption, J01A tetracyclines, J01B amphenicols, J01C penicillins, J01D other beta-lactams, J01E sulfonamides and trimethoprim, J01F macrolides, lincosamides & streptogramins, J01G aminoglycosides, J01M quinolones	Global antimicrobial resistance and use surveillance system (GLASS)	https://www.who.int/initiatives/glass
	European Surveillance of Antimicrobial Consumption Network (ESAC-Net)	https://www.ecdc.europa.eu/en/antimicrobial-consumption/surveillance-and-disease-data/database
	Global Research on Antimicrobial Resistance (GRAM) project	https://www.tropicalmedicine.ox.ac.uk/gram/research/visualisation-app-antibiotic-usage-and-consumption
Income Groups	World Bank Country and Lending Groups	https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups
Animal antimicrobial consumption, aminoglycosides, amphenicols, cephalosporins (all generations), lincosamides, macrolides, fluoroquinolones, other quinolones, penicillins, pleuromutilins, polypeptides, sulfonamides (including trimethoprim), tetracyclines, others	ANImal antiMicrobial USE Global Database (ANIMUSE)	https://amu.woah.org/amu-system-portal/home

* CAESAR network is part of a joint initiative including the WHO Regional Office for Europe (WHO/Europe), European Society of Clinical Microbiology and Infectious Diseases (ESCMID) and Dutch institute for public health and the environment (RIVM).

Table S4 Description of scenarios

Scenario	Definition
Baseline	Linearly inference of socio-economic variables based on the trend of past data.
Sustainable development	By 2030, universal access to WASH services, comprehensive immunization coverage, health investments conducive to sustainable development (over 5% GDP), and preventing catastrophic household health expenditure due to high out-of-pocket costs (less than 20% health expenditures).
Antimicrobial consumption cutdown	The consumption of antimicrobials decreases to 50% of the baseline scenario.
SSP1-2.6	SSP1-2.6 merges a sustainable development pathway (SSP1) with a low greenhouse gas concentration trajectory (RCP2.6), envisioning a future oriented towards sustainability, with substantial investments in education, health, and renewable energy, targeting low emissions and striving to limit global warming to well below 2°C. This pathway emphasizes environmental protection and social equity.
SSP2-4.5	SSP2-4.5 pairs a moderate development pathway (SSP2) with an intermediate greenhouse gas concentration trajectory (RCP4.5), depicting a future that follows historical trends, with moderate economic development and emissions, and partial progress towards sustainability, but lacking radical behavioural shifts or major advances in green technology.
SSP3-7.0	SSP3-7.0 links a fragmented world scenario (SSP3) with a high-emission trajectory (RCP7.0), portraying a future of regional disputes, prioritizing national security over global environmental concerns, leading to high emissions and scant international cooperation on climate change, potentially resulting in significant global warming and adaptation challenges.
SSP5-8.5	SSP5-8.5 combines a fossil-fuel intensive development pathway (SSP5) with the highest greenhouse gas concentration trajectory (RCP8.5), imagining a future driven by rapid economic growth reliant on fossil fuels, resulting in high energy consumption and emissions, a scenario that could lead to severe climate change impacts due to extremely high global temperatures.

* WASH = Water, Sanitation, and Hygiene. RCP = Representative Concentration Pathway.